

Akhil Kora

Buffalo Grove, IL • akhilkora@gmail.com • (224)-584-6097

OBJECTIVE

Tenth-grade student on a competitive robotics team with hands-on experience in CAD, Arduino, and programming. Seeking a summer robotics internship or lab-assistant role to build practical engineering skills and contribute reliably to a team.

EDUCATION

Adlai E. Stevenson High School — Buffalo Grove, IL Expected Graduation June 2029

Grade 10 • GPA: 4.0/4.14

Relevant Coursework: Computer Science, Algebra II, Intro to Engineering & Design (CAD), Chemistry

ROBOTICS EXPERIENCE

FIRST Tech Challenge — Team #24108 “Boltic” June 2026 – Present

Design & Hardware Subteams

- Designed a custom coaxial swerve drivetrain using Fusion360 to improve steering and holonomic motion on the field.
- Built/assembled a custom parallel plate chassis, and a custom turret as I learned about CNC manufacturing, 3D printing, and basic soldering.

PROJECTS

Custom Coaxial Swerve Drivetrain Fusion360 • FTC

- Built a two-motor robot with infrared sensors that tracks a taped course; tuned a proportional controller to reduce overshoot on turns.
- Wired and soldered the motor driver and sensor array from scratch.

Rotating Turret Assembly Fusion • CNC

- Custom turret fabricated using CNC machinery and 3D prints
- Adjustable turret hood to alter the trajectory of the ball
- OpenCV camera to track the target

Parallel Plate Chassis Fusion • CNC

- Lightweight parallel robot frame
- Modular for quick mechanical fixes

Artzy Python • Streamlit

- Web app built using Python and Streamlit
- Converts images/photos into clean line-art sketches so that people can color in

SafeSea React • Firebase

- Web app built using React and firebase with typescript as the primary language
- Helps prevent collisions between sailors and divers by tracking both

Major Odyssey React • Firebase

- web app built using react and firebase with typescript as the primary language
- Lets students experience the major they're committing to before college through immersive, game-like simulations

Energy Assist FlutterFlow • API's

- Designed using FlutterFlow and dart is it's primary language

- Helps you manage and track your energy usage throughout the day, and cut down your electricity bill by shifting appliance use to low-price hours.

SKILLS

CAD & Fabrication: Fusion 360, 3D printing, basic soldering

Programming: Python, C ++, Git, Typescript

Electronics: Arduino, DC/servo motors, sensors (IR, Gyro, ultrasonic, encoders), breadboarding

Tools: Git basics, Google Workspace, hand and power tools (supervised)

ACTIVITIES & LEADERSHIP

- STEM Outreach — demoed team robots at a local middle school for ~250 students.
- Design Lead, Robotics — created team robot design to compete in the First Tech Challenge(FTC).
- Science Fair — competed at state for fluid dynamics, a subcategory of physics.
- Non-Profit — Formed a non-profit to educate middle schoolers about Patent Law and Intellectual Property

HONORS

- Honor Roll — two semesters.
- FIRST Tech Challenge — Design Award winner (regional, 2026).
- Science Fair — Best in category award winner and State Qualifier (State, 2025).
- FIRST Lego League — Robot Design 4th place award winner (national, 2025).
- FIRST Lego League — Champions Award 3rd place award winner (state, 2024).
- FIRST Lego League — Robot Design 1st place award winner (national, 2024).
- FIRST Lego League — Robot Performance 3rd place award winner (national, 2024).
- FIRST Lego League — Robot Performance 2nd place award winner national qualifier (national, 2023).
- FIRST Lego League — Champions Award 2nd place award winner national qualifier (national, 2023).
- Startups in Business and Tech Club — 2nd Stage qualification (school, 2026).